

▼ **Figure 56.3 A hundred heartbeats from extinction.** These are two members of what Harvard biologist E. O. Wilson calls the Hundred Heartbeat Club, species with fewer than 100 individuals remaining on Earth. The Yangtze River dolphin may be extinct, but a few individuals were reportedly sighted in 2007.

Philippine eagle



Yangtze River dolphin



? To document that a species has actually become extinct, what factors would you need to consider?

- In North America, at least 123 freshwater animal species have become extinct since 1900, and hundreds more species are threatened. The extinction rate for North American freshwater fauna is about five times as high as that for terrestrial animals.

The local populations of a species can also be driven to extinction; for example, populations of a species may be lost in one river system but survive in an adjacent one. Global extinction of a species means that it is lost from *all* the ecosystems in which it lived, leaving them permanently impoverished.

➔ **Mastering Biology HHMI Video: Surveying Gorongosa's Biodiversity**



Ecosystem Diversity

The variety of ecosystems on Earth is a third level of biological diversity. Because of the many interactions between members of different species in an ecosystem, the extinction of populations of one species can have a negative impact on other species in the ecosystem (see Figure 54.20). For instance, bats called “flying foxes” are important pollinators and seed dispersers in the Pacific Islands, where they are increasingly hunted as a luxury food (**Figure 56.4**). Conservation biologists fear that the extinction of flying foxes would also harm the native plants of the Samoan Islands, where four-fifths of the tree species depend on flying foxes for pollination or seed dispersal.

▼ **Figure 56.4 The endangered Marianas “flying fox” bat (*Pteropus mariannus*), an important pollinator.**



Some ecosystems have already been heavily affected by humans, and others are being altered at a rapid pace. Since European colonization, more than half of the wetlands in the contiguous United States have been drained and converted to agricultural and other uses. In California, Arizona, and New Mexico, roughly 90% of native riparian (streamside) communities have been affected by overgrazing, flood control, water diversions, lowering of water tables, and invasion by non-native plants.

Biodiversity and Human Welfare

Why should we care about the loss of biodiversity? One basic reason concerns our human sense of connection to nature and all forms of life, termed *biophilia*. Moreover, the belief that other species are entitled to life is a pervasive theme of many religions and the basis of a moral argument that we should protect biodiversity. There is also a concern for future human generations. Paraphrasing an old proverb, G. H. Brundtland, a former prime minister of Norway, said, “We must consider our planet to be on loan from our children, rather than being a gift from our ancestors.” In addition to such philosophical and moral justifications, species and genetic diversity bring us many practical benefits.

Benefits of Species and Genetic Diversity

Many threatened species could potentially provide medicines, food, and fibers for human use, making biodiversity a crucial natural resource. Products from aspirin to antibiotics were derived originally from natural sources. If we lose wild populations of plants closely related to agricultural species, we lose genetic resources that could be used to improve crop qualities, such as disease resistance. For instance, in the 1970s, plant breeders responded to devastating outbreaks of the grassy stunt virus in rice (*Oryza sativa*) by screening